

Foundation Mathematics 1017SCG
Week 4 Workshop Questions

1. Expand and simplify the following.

(a) $3(x + 6)$

(b) $2(2y + 4)$

(c) $5x(2x - 3)$

(d) $4y(5y - 1)$

(e) $7x(2x + 3) - 5x$

(f) $-3y(5y - 2) + 4$

(g) $-2(n - 4) - (3n + 1)$

(h) $x(2x - 3) + 4(x^2 + 2)$

(i) $2a(5a - 1) + 3(4a + 2)$

(j) $2g(1 - 2g) - 5(g - 2)$

(k) $4(x - 3) - 2(x - 2)$

(l) $12(n - 3) + n(-2 - 2n)$

2. Expand and simplify the following.

(a) $(a + 4)^2$

(b) $(2x - 3)^2$

(c) $(y - 7)^2$

(d) $(n + 6)(n - 6)$

(e) $(y + 2)(y - 2)$

(f) $(3x + 4)(3x - 4)$

3. Expand and simplify the following.

(a) $(x + 2)(x + 3)$

(b) $(x + 3)(x + 8)$

(c) $(x - 4)(x - 5)$

(d) $(2y + 7)(4y - 1)$

(e) $(3b + 8)(10b - 3)$

(f) $(7h - 2)(3h - 5)$

(g) $(a - 2)(a + 5)$

(h) $(2x - 1)(3x + 4)$

4. Factorise the following.

(a) $6x + 2$

(b) $5x - 10$

(c) $x^2 - 8x$

(d) $2x^3 + 7x^2$

(e) $6y^2 + 9y$

(f) $10y^5 - 25y^3$

(g) $4a^3 + 12a^2$

(h) $2x^3y^2 - 4x^3y^5$

(i) $14x^2y^2 + 21x^4y$

(j) $3a^2b - 15ab^2$

5. Factorise the following.

(a) $x^2 - 25$

(b) $y^2 - 9$

(c) $x^2 - 16$

(d) $y^2 - 1$

(e) $4x^2 - 81$

(f) $100y^2 - 49$

(g) $5x^2 - 20$

(h) $3y^2 - 75$

6. Factorise the following.

(a) $x^2 + 5x + 6$

(b) $y^2 + 11y + 10$

(c) $x^2 + 6x + 8$

(d) $y^2 - 5y + 4$

(e) $x^2 - 13x + 12$

(f) $a^2 - 11a - 12$

(g) $y^2 - 9y + 20$

(h) $2n^2 + 14n + 20$

(i) $3k^2 + 33k + 30$

(j) $-x^2 - 8x - 12$

7. Factorise the following.

(a) $2x^2 - 11x + 5$

(b) $2x^2 + 5x + 3$

(c) $3x^2 + 14x + 8$

(d) $3x^2 - 8x + 5$

(e) $4x^2 + 9x + 2$

(f) $4x^2 - 8x + 3$

8. Simplify the following. (Hint: If possible, start by factoring)

(a) $\frac{x}{25} \times \frac{10}{y}$

(b) $\frac{20}{21a} \times \frac{7c}{10}$

(c) $\frac{3nm}{5} \div \frac{9n}{10}$

(d) $\frac{x-7}{1-x^2} \div \frac{2x-14}{x+1}$

(e) $\frac{3y}{2y^2 - y - 10} \times \frac{2y-5}{9y^2}$

(f) $\frac{4a}{a^2 + a - 6} \div \frac{2a}{a^2 - 9}$

(g) $\frac{2x^2 - 5x - 3}{x^2 - 4} \div \frac{x^2 - 3x}{2x + 4}$

9. (a) Expand $(2x + 3)(x - 4)$.

(b) Using your answer from part (a), expand $(2x + 3)(x - 4)(x + 1)$.